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Comprehensive Question Bank: Unit 7 — Legal and Ethical Aspects of Computing System

Section A: Multiple-Choice Questions (MCQs)

Topic 7.1.1: Importance of Terms of Use

1. What is the primary purpose of "Terms of Use" agreements?
A) To advertise new products to users
B) To establish a fair and transparent relationship between provider and user
C) To increase a company's stock price
D) To replace a user's need for a password
2. Which right is protected by Terms of Use, according to the text?
A) The right to unlimited free service
B) The right to privacy
C) The right to ignore company policies
D) The right to sell the company's data
3. A Pakistani online education platform includes terms prohibiting the sharing of course content without permission. This is an example of which importance category of Terms of Use?
A) Clarity and Transparency
B) Legal Safeguards
C) Personal Rights
D) Ethical Considerations
4. Why do Terms of Use for a food delivery service like Foodpanda specify what happens if an order is incorrect or delayed?
A) To increase delivery fees
B) To provide clarity and transparency to prevent misunderstandings
C) To advertise other restaurants
D) To violate consumer rights
5. What best describes the "Legal Safeguards" function of Terms of Use for a business?
A) They guarantee the company will never make mistakes
B) They set boundaries that protect the company from legal issues caused by user misuse
C) They eliminate the need for customer service
D) They allow the company to ignore user complaints

Topic 7.1.2: Common Clauses and Conditions

1. Which clause requires users of a ride-sharing app like Careem to provide a valid phone number?
A) Limitations of Liability
B) User Obligations
C) Termination of Service
D) Intellectual Property Rights
 2. If an online banking app is temporarily down, which clause typically limits the bank's responsibility for the disruption?
A) User Obligations
B) Limitations of Liability
C) Privacy and Data Use
D) Intellectual Property Rights
 3. Which clause explains how a company like WhatsApp stores and protects user messages?
A) Termination of Service
B) User Obligations
C) Privacy and Data Use
D) Intellectual Property Rights
 4. A Pakistani news website prohibits users from copying and distributing its articles without permission. This is protected under which clause?
A) User Obligations
B) Limitations of Liability
C) Intellectual Property Rights
D) Privacy and Data Use
 5. Facebook's ability to suspend accounts that spread hate speech or misinformation falls under which clause?
A) Privacy and Data Use
B) User Obligations
C) Termination of Service
D) Limitations of Liability
 6. Which of the following is NOT one of the five common clauses discussed in Terms of Use?
A) User Obligations
B) Intellectual Property Rights
C) Product Advertising
D) Termination of Service
-

Topic 7.1.3: Legal Ramifications

1. What does the term "ramifications" mean in the context of violating Terms of Use?
A) Rewards
B) Consequences
C) Suggestions
D) Discounts
 2. What legal consequence might a user face for using software without a proper license?
A) A free upgrade
B) Legal action for software piracy
C) A loyalty reward
D) Automatic account verification
 3. Besides legal risk, what additional danger do companies using unlicensed software commonly face?
A) Increased customer satisfaction
B) Cybersecurity threats
C) Lower operating costs
D) Guaranteed profit growth
-

Topic 7.1.4: Ethical Considerations

1. Which three principles are highlighted as key ethical considerations in Terms of Use?
A) Speed, cost, and popularity
B) Fairness, transparency, and respect for user rights
C) Advertising, branding, and marketing
D) Profit, growth, and expansion
2. When a Pakistani online course platform clearly specifies whether a certificate will be awarded, how data will be handled, and if there are hidden fees, it is demonstrating which ethical principle?
A) Legal ramification
B) Transparency
C) Termination of service
D) Intellectual property protection
3. A free app's Terms of Use state it owns all uploaded content and may use it to train AI models. Even if legal, what ethical question does this raise?
A) Whether the app has a good user interface
B) Whether users truly understand and fairly consent to what they are giving up
C) Whether the app is expensive
D) Whether the app requires an internet connection

Topic 7.1.5: Personal Rights

1. Which of the following is listed as a personal right within Terms of Use?
 - A) The right to unlimited data storage
 - B) The right to withdraw consent**
 - C) The right to ignore updates
 - D) The right to sell company property
 2. Under Pakistan's Personal Data Protection Bill, what right do users have regarding their personal data?
 - A) The right to sell it to third parties
 - B) The right to access it and request corrections or deletions**
 - C) The right to prevent all companies from operating
 - D) The right to demand free products
 3. Why is it recommended that users periodically review a platform's Terms of Use?
 - A) Because platforms remove Terms of Use permanently
 - B) Because platforms frequently update their Terms of Use**
 - C) Because it is legally required every day
 - D) Because Terms of Use expire after one use
-

Topic 7.2.1.1: Spam

1. What is "spam" in the context of digital communication?
 - A) A type of harmful spyware
 - B) Unwanted messages received via email or phone**
 - C) A file stored by a website on your device
 - D) A technique for redirecting users to fake websites
 2. Which of the following is the best example of spam?
 - A) A virus secretly recording your keystrokes
 - B) Repeated text messages from unknown numbers advertising unrequested products**
 - C) A fake bank website replacing the real one
 - D) A cookie remembering your login details
-

Topic 7.2.1.2: Spyware

1. What does spyware do once installed on a device?
 - A) It speeds up the device's performance
 - B) It secretly monitors user activity without their knowledge**
 - C) It automatically deletes unwanted emails
 - D) It blocks all internet access
 2. A "free flashlight app" requests access to your contacts, microphone, and location. What should this immediately suggest?
 - A) The app is highly rated
 - B) The app may contain spyware, since these permissions are unrelated to its function**
 - C) The app will work faster
 - D) The app is from a verified trusted source
 3. Downloading a free app from an unreliable source poses what specific privacy risk, according to the text?
 - A) It may charge a hidden subscription fee
 - B) It may install spyware that secretly monitors your online activities**
 - C) It may improve your device's battery life
 - D) It may automatically report spam to authorities
-

Topic 7.2.1.3: Cookies

1. What is the primary function of a website cookie?
 - A) To infect a device with a virus
 - B) To help a website remember information like login details and preferences**
 - C) To permanently delete browsing history
 - D) To redirect users to a fraudulent website
2. Which type of cookie is set by a domain different from the website a user is directly visiting, and is commonly used for tracking across multiple sites?
 - A) First-party cookie
 - B) Third-party cookie**
 - C) Session-only cookie
 - D) Anti-malware cookie
3. Which of the following is a typical use of a first-party cookie?
 - A) Tracking a user's behavior across unrelated websites for advertisers
 - B) Remembering a user's login details or shopping cart on the site they are visiting**
 - C) Installing spyware on the user's device
 - D) Redirecting the user to a fake banking website

4. Compared to first-party cookies, third-party cookies generally carry:
- A) Lower privacy risk
 - B) Higher privacy risk, since they build a profile across multiple websites**
 - C) No privacy risk at all
 - D) The same risk level in every case
-

Topic 7.2.1.4: Phishing and Pharming

1. What is "phishing"?
 - A) A technique that hijacks the internet's address system
 - B) A scam where someone pretends to be a trustworthy organization to trick users into giving personal information**
 - C) A file stored on a device by a website
 - D) Unwanted advertising messages
2. According to the text, what percentage of phishing attacks are carried out through email?
 - A) Over 50%
 - B) Over 70%
 - C) Over 90%**
 - D) Exactly 100%
3. What makes "pharming" more dangerous than phishing in most cases?
 - A) It only works on mobile phones
 - B) It can redirect users to a fake website even when they type the correct web address**
 - C) It requires the user to open an email attachment
 - D) It only targets business email accounts
4. Which technique involves attackers corrupting the "internet's phonebook" (DNS) to secretly redirect users?
 - A) Phishing
 - B) Spam
 - C) Spyware
 - D) Pharming**
5. A user receives an email claiming to be from their bank, asking them to click a link and enter account details. This is an example of:
 - A) Pharming
 - B) Phishing**
 - C) Spyware
 - D) A first-party cookie

6. What is the key difference between phishing and pharming?
- A) Phishing hijacks DNS servers; pharming uses deceptive emails
 - B) Phishing uses a deceptive message to trick the user; pharming hijacks the address system itself, requiring no mistake from the user**
 - C) They are exactly the same technique
 - D) Phishing only affects businesses; pharming only affects individuals
-

Topic 7.2.2.1: Spam Filters

1. What is the main function of a spam filter?
 - A) To install antivirus software automatically
 - B) To keep unwanted emails out of a user's inbox**
 - C) To redirect users to fake websites
 - D) To manage website cookies
 2. If a user receives many unwanted promotional emails, what tool can automatically move them to a separate folder or delete them?
 - A) A cookie manager
 - B) A spam filter**
 - C) A pharming detector
 - D) An intellectual property clause
-

Topic 7.2.2.2: Spyware Protection (Anti-Malware & Permissions)

1. What is the function of antivirus/anti-malware software?
 - A) To generate Terms of Use documents
 - B) To protect a computer from harmful programs like spyware by scanning files for threats**
 - C) To create third-party cookies
 - D) To redirect DNS requests
2. When auditing app permissions, what is the key question to ask about each permission an app requests?
 - A) "Does this permission make the app faster?"
 - B) "Does this app actually need this permission to function?"**
 - C) "Is this permission free to use?"
 - D) "Does this permission increase the app's rating?"
3. In an app permission audit, why would a weather app's request for "Location" access be considered reasonable, while its request for "Contacts" access would not?
 - A) Because location access is always risky and contacts access is always safe

- B) Because location matches the app's core function (local forecasts), while contacts does not**
- C) Because weather apps never need any permissions
- D) Because contacts access improves weather accuracy
-

Topic 7.2.2.3: Cookie Management

1. What does "cookie management" involve?
A) Deleting all files from a computer
B) Deciding which cookies to keep and which ones to delete
C) Creating new Terms of Use clauses
D) Installing antivirus software
 2. Which browser setting can a user enable to prevent advertisers from tracking them across multiple websites?
A) Enable third-party cookies
B) Block third-party cookies
C) Disable spam filters
D) Enable pharming protection
-

Topic 7.2.2.4: Recognizing Phishing

1. In a phishing email analysis, what is a major red flag when the sender's domain is "yourbank-secure-verify.com" instead of the bank's actual domain?
A) It indicates the email is completely safe
B) It suggests domain spoofing, a common phishing tactic
C) It means the email was sent from Pakistan
D) It confirms the email is from a trusted source
2. Why do phishing emails often use urgent language like "Your account will be suspended in 24 hours"?
A) To provide helpful information calmly
B) To create pressure so the victim acts quickly without thinking
C) To comply with legal requirements
D) To improve customer service ratings
3. What is the safest way to verify a suspicious email claiming to be from your bank?
A) Click the link provided in the email immediately
B) Independently type the bank's known website address or call their official support

number

- C) Reply to the email with your account details
 - D) Forward the email to friends for their opinion
-

Topic 7.2.2.5: Guarding Against Pharming

1. What two visual indicators should a user check in the browser address bar to guard against pharming?
 - A) A large logo and a bright color
 - B) "https://" and a padlock symbol**
 - C) The number of ads on the page
 - D) The website's loading speed
 2. What does clicking the padlock icon in a browser reveal, which helps guard against pharming?
 - A) The website's advertising budget
 - B) The site's security certificate details, such as who it was issued to**
 - C) The number of cookies stored
 - D) The website's spam filter settings
-

Topic 7.3.1: Causes of the Digital Divide

1. Which of the following is described as an economic barrier contributing to the digital divide?
 - A) Users choosing not to use the internet for fun
 - B) Families being unable to afford computers, smartphones, or broadband internet**
 - C) Websites having too many features
 - D) Governments banning all internet use
2. What kind of barrier is present when remote or rural areas lack the infrastructure necessary for internet connectivity?
 - A) Economic barrier
 - B) Geographical barrier**
 - C) Social barrier
 - D) Legal barrier
3. A person has access to a computer and internet but does not know how to use them effectively. This illustrates which cause of the digital divide?
 - A) Geographical barrier
 - B) Economic barrier
 - C) Educational barrier (lack of digital literacy)**
 - D) Termination of service

4. According to the text, which factors are identified as social barriers affecting access to technology?
- A) Price and speed
 - B) Age, gender, and disability**
 - C) Software version and hardware brand
 - D) Time zone and climate
-

Topic 7.3.2.1–7.3.2.5: Impacts of the Digital Divide

1. During the COVID-19 pandemic, which group struggled most to keep up with online learning, according to the text?
 - A) Students in urban private schools with reliable internet
 - B) Students from low-income families or rural areas lacking internet access or devices**
 - C) Teachers working from school offices
 - D) Students who already owned laptops
2. What economic consequence can result from a lack of access to online job portals and digital banking services?
 - A) Higher employment rates for rural populations
 - B) Higher unemployment rates in disadvantaged communities**
 - C) Lower prices for digital devices
 - D) Increased access to e-government services
3. How does the digital divide affect social and civic participation, based on the Pakistan example given?
 - A) Rural populations have more access to e-government services than urban populations
 - B) Rural populations have limited means to participate in governmental processes due to unequal access**
 - C) Civic participation is unaffected by internet access
 - D) E-government services are banned nationwide
4. Which impact of the digital divide relates specifically to telemedicine and health education access?
 - A) Digital Literacy Gap
 - B) Economic Disparities
 - C) Health Disparities**
 - D) Educational Inequality
5. What does the "Digital Literacy Gap" impact describe?
 - A) A gap in internet pricing between regions
 - B) A gap in digital skills that reinforces the divide between tech-savvy and non-tech-**

savvy populations

- C) A gap in the number of available websites
 - D) A gap in government tax revenue
-

Topic 7.3.3: Bridging the Digital Divide

1. Which of the following is listed as a way to bridge the digital divide?
 - A) Restricting internet access further
 - B) Government initiatives to build necessary infrastructure**
 - C) Ignoring rural communities
 - D) Removing all educational programs
 2. What is the purpose of "Public-Private Partnerships" in bridging the digital divide?
 - A) To make technology more expensive
 - B) To combine government, business, and nonprofit efforts to make technology more affordable**
 - C) To eliminate government involvement entirely
 - D) To restrict technology to urban areas only
 3. What initiative has the Government of Pakistan launched to increase internet access in rural areas?
 - A) Banning smartphones in rural areas
 - B) Introducing mobile broadband services in remote regions**
 - C) Removing all public schools
 - D) Increasing internet taxes
-

Topic 7.4.1: Positive Impacts of Computing

1. How have platforms like Coursera and edX positively impacted education?
 - A) By restricting access to only wealthy students
 - B) By allowing students from remote areas to take courses from prestigious universities worldwide**
 - C) By eliminating the need for teachers entirely
 - D) By making education more expensive
2. Which social movement is mentioned as having gained global traction partly through social media?
 - A) The Industrial Revolution
 - B) The #MeToo movement**
 - C) The printing press revolution
 - D) The invention of the telephone

3. How did social media contribute to events surrounding the 2014 Peshawar school attack in Pakistan?
- A) It suppressed all information about the event
 - B) It played a crucial role in raising awareness and mobilizing support and solidarity**
 - C) It had no impact on public awareness
 - D) It caused the attack to be forgotten quickly
4. How has computing driven economic growth in Pakistan, according to the text?
- A) By eliminating all traditional businesses
 - B) By enabling e-commerce platforms like Daraz to help small businesses reach nationwide and international customers**
 - C) By increasing unemployment
 - D) By restricting business growth to urban centers only
-

Topic 7.4.2: Negative Impacts of Computing

1. What major negative impact is associated with the collection and storage of vast amounts of personal data without proper regulation?
- A) Improved communication
 - B) Privacy breaches and data misuse**
 - C) Increased digital literacy
 - D) Reduced misinformation
2. Which real-world Pakistani incident is cited as an example of a privacy/data protection concern?
- A) The introduction of mobile broadband
 - B) The leakage of citizens' data from NADRA**
 - C) The launch of Coursera
 - D) The rise of Daraz
3. What negative effect did misinformation about COVID-19 have on the public, according to the text?
- A) It had no measurable effect
 - B) It caused confusion and fear among the public**
 - C) It improved public health outcomes
 - D) It eliminated the need for vaccines
4. What is identified as a necessary skill for people to critically evaluate information they encounter online?
- A) Coding skills

- B) Digital literacy**
 - C) Financial literacy
 - D) Graphic design skills
-

Topic 7.5.1: Responsible Digital Behavior

1. Which of the following is an example of a strong password practice mentioned in the text?
 - A) Using "password123"
 - B) Using a combination of letters, numbers, and symbols**
 - C) Using your own name only
 - D) Reusing the same simple password everywhere
 2. What should a user do if they receive an email claiming to be from a bank asking for account details?
 - A) Immediately reply with the requested information
 - B) Verify its authenticity before responding**
 - C) Forward it to as many people as possible
 - D) Ignore all future emails from that bank
-

Topic 7.5.2.1–7.5.2.2: Ethical Use of Information

1. What does "responsible data sharing" involve?
 - A) Sharing personal information with any website that asks
 - B) Only sharing personal or sensitive information when necessary and with trusted entities**
 - C) Refusing to ever share any information
 - D) Sharing information only on social media
2. What is "plagiarism"?
 - A) Properly citing sources in an assignment
 - B) Using someone else's ideas or words without proper attribution**
 - C) Creating original research
 - D) Sharing data responsibly
3. What ethical issue arises when a computing system is trained on data that reflects existing human prejudices?
 - A) Improved system performance
 - B) Data bias, where the system learns and repeats unfair patterns**
 - C) Increased digital literacy
 - D) Enhanced accessibility

4. According to the text, what should students do when writing a research paper that includes ideas from an online article?
- A) Copy the text directly without any changes
 - B) Cite the source instead of copying text without attribution**
 - C) Avoid using any outside sources
 - D) Only use information found on social media
-

Topic 7.5.3: Cybersecurity Awareness

1. Where can a cybercrime be reported in Pakistan, according to the text?
- A) A local newspaper
 - B) NR3C-FIA (<https://www.nr3c.gov.pk>)**
 - C) A private security company only
 - D) A social media platform's help center
2. Cybersecurity awareness is best described as:
- A) A one-time software installation
 - B) The ongoing practice of understanding and implementing protections against online threats**
 - C) A legal document signed once
 - D) A feature exclusive to antivirus software
-

Topic 7.5.4.1–7.5.4.2: Collaborative Problem Solving

1. What does "human-machine collaboration" involve?
- A) Replacing all human decision-making with machines
 - B) Working together with technology, combining human judgment with machine capabilities**
 - C) Avoiding the use of any digital tools
 - D) Using only human effort without any software
2. In a group science project, students collaborating and sharing research findings is an example of:
- A) Human-machine collaboration
 - B) Teamwork and knowledge sharing**
 - C) Data bias
 - D) Pharming prevention

3. Why is human judgment still important even when using powerful AI tools for a group project?
- A) Because AI tools are always completely accurate
 - B) Because blindly trusting AI output without verification can lead to errors**
 - C) Because human judgment slows down every project
 - D) Because AI tools cannot process any data
-

Topic 7.5.5.1–7.5.5.2: Creating Accessible Digital Content

1. What is the purpose of "ALT text" for images on a webpage?
- A) To make images load faster
 - B) To provide descriptions so visually impaired users relying on screen readers can understand the content**
 - C) To increase the image's file size
 - D) To replace the need for headings
2. Why is color contrast important in accessible digital design?
- A) It has no impact on readability
 - B) It ensures text is readable for users with low vision**
 - C) It only matters for printed documents
 - D) It is required only for video content
3. What is the recommended minimum contrast ratio mentioned in the text for normal text to meet accessibility guidelines?
- A) 1:1
 - B) 2:1
 - C) 4.5:1**
 - D) 10:1
4. What is the purpose of gathering "user feedback" for digital content?
- A) To avoid ever making changes to a website
 - B) To collect opinions from real users and make necessary improvements**
 - C) To replace the need for accessibility features
 - D) To increase advertising revenue
5. Structuring headings in a logical order (Heading 1, Heading 2, Heading 3) on a webpage primarily helps:
- A) Increase the page's loading speed
 - B) Screen readers announce the page's structure logically to visually impaired users**
 - C) Improve the page's color scheme
 - D) Reduce the number of images needed
-

Section B: Short Answer Questions (SQs)

Topic 7.1: Understanding Terms of Use

1. Define "Terms of Use" in your own words.
2. Explain why understanding Terms of Use is important for protecting user rights.
3. Differentiate between the "Legal Safeguards" and "Clarity and Transparency" purposes of Terms of Use.
4. List three common clauses found in Terms of Use agreements.
5. Explain the purpose of a "Limitations of Liability" clause with an example.
6. What legal ramifications might a user face for violating Terms of Use?
7. Explain the ethical considerations companies should keep in mind when writing Terms of Use.
8. Identify two personal rights users have under Terms of Use agreements.
9. Why is it recommended to periodically review a platform's Terms of Use?
10. A free app's Terms of Use state it can use uploaded photos to train AI models. Briefly explain what rights the user may be giving up.

Topic 7.2: Privacy and Security Threats

1. Define "spam" and provide one example.
2. Explain how spyware can end up on a user's device without their knowledge.
3. Differentiate between first-party cookies and third-party cookies.
4. Define "phishing" and describe one warning sign that identifies it.
5. Differentiate between phishing and pharming.
6. Explain why pharming is considered more dangerous than phishing for an average user.
7. Briefly explain how a spam filter works to protect a user's inbox.
8. Explain the role of antivirus/anti-malware software in preventing spyware infections.
9. State the steps you would take to audit app permissions on a mobile device.
10. Explain how a user can guard against pharming when visiting a banking website.
11. List two red flags that help identify a phishing email.
12. Explain why an app requesting permissions unrelated to its core function is a security concern.

Topic 7.3: The Digital Divide and Its Impacts

1. Define the term "digital divide."
2. List the four causes of the digital divide mentioned in the text.
3. Explain how economic barriers contribute to the digital divide.
4. Differentiate between geographical barriers and social barriers as causes of the digital divide.

5. Explain how the digital divide creates educational inequality, using the COVID-19 example.
6. Identify two impacts of the digital divide on social and civic participation.
7. Explain how the digital divide affects healthcare access.
8. Define the "Digital Literacy Gap" and explain how it reinforces existing inequality.
9. List three strategies for bridging the digital divide.
10. Briefly explain one Pakistani government initiative aimed at reducing the digital divide.

Topic 7.4: Computing's Impact on Individuals and Society

1. List two positive impacts of computing systems on society.
2. Explain how e-commerce platforms like Daraz have impacted small businesses in Pakistan.
3. Identify two negative impacts of computing on individuals and society.
4. Explain how misinformation spread during the COVID-19 pandemic, and why digital literacy is important in response.
5. Briefly describe the NADRA data leak example and what it illustrates about privacy concerns.
6. Explain how social media contributed to raising awareness after the 2014 Peshawar school attack.

Topic 7.5: Digital Citizenship and Ethical Considerations

1. Define "digital citizenship" in your own words.
 2. List two examples of responsible digital behavior.
 3. Differentiate between "responsible data sharing" and general data sharing.
 4. Define "plagiarism" and explain how to avoid it.
 5. Explain what is meant by "data bias" in computing systems.
 6. Where can a cybercrime be reported in Pakistan?
 7. Differentiate between human-machine collaboration and teamwork/knowledge sharing.
 8. Explain the purpose of ALT text in accessible digital design.
 9. State two design elements that make digital content more accessible.
 10. Explain why gathering user feedback is important for accessible digital content.
 11. Explain why human judgment remains necessary even when using AI tools for a group project.
-

Section C: Long / Extensive Questions (LQs)

Topic 7.1: Understanding Terms of Use

1. Explain the common clauses found in Terms of Use and describe how they protect both the service provider and the user.
 - a) Define "Terms of Use" and explain its overall purpose.
 - b) Describe at least four common clauses (e.g., User Obligations, Limitations of Liability, Privacy and Data Use, Intellectual Property Rights, Termination of Service), providing a real-world example for each.
 - c) Discuss the legal ramifications a user may face for violating these terms.
 - d) Evaluate whether current Terms of Use practices adequately protect user personal rights, using examples from Pakistan's Personal Data Protection Bill.
 2. Analyze the relationship between legal compliance and ethical responsibility in Terms of Use agreements.
 - a) Explain the difference between something being "legal" and something being "ethical" in the context of a company's Terms of Use.
 - b) Using the example of a free photo-editing app that claims ownership of uploaded photos, evaluate whether this practice is ethical even if it is legal.
 - c) Propose two changes a company could make to its Terms of Use to be more transparent and fair to users.
-

Topic 7.2: Privacy and Security Threats

1. Discuss in detail the major types of privacy and security threats covered in this unit, and explain corresponding prevention techniques for each.
 - a) Define and explain spam, spyware, cookies, phishing, and pharming, providing one real-world example for each.
 - b) For each threat identified in part (a), describe at least one specific prevention or mitigation technique.
 - c) Construct a step-by-step guide a student could follow to identify a phishing email, including at least four distinct red flags.
 - d) Evaluate why pharming is considered a more advanced and dangerous threat than phishing, and justify why traditional user caution (like checking spelling) is insufficient against it.
2. Analyze and compare cookies, spyware, and phishing as distinct privacy risks.
 - a) Construct a comparison table evaluating each threat by: how it operates, what information it targets, and how a user can protect against it.
 - b) Discuss the difference between first-party and third-party cookies, and evaluate the privacy

- trade-offs of blocking third-party cookies entirely.
- c) Design a complete mobile device permission-auditing workflow that a non-technical user could follow to reduce their risk of spyware.
3. Evaluate the importance of combining multiple security strategies rather than relying on a single technique.
- a) Explain why spam filters, antivirus software, cookie management, and URL verification are each necessary but individually insufficient.
- b) Construct a real-world scenario in which a user is protected from one threat (e.g., phishing) but still falls victim to another (e.g., pharming), and explain why.
- c) Justify why combining multiple prevention techniques provides significantly stronger protection than any single technique alone.
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Topic 7.3: The Digital Divide and Its Impacts

1. Explain the concept of bridging the digital divide and discuss the roles of government initiatives in addressing this issue. Provide relevant examples from Pakistan.
- a) Define the digital divide and explain its four main causes.
- b) Discuss at least three impacts of the digital divide (e.g., educational, economic, social/civic, health, digital literacy) in detail, with examples.
- c) Explain the roles of government initiatives, educational programs, and public-private partnerships in bridging the digital divide.
- d) Evaluate the effectiveness of Pakistan's mobile broadband initiatives in addressing rural connectivity gaps, and suggest one additional strategy that could further reduce the divide.
2. Analyze the compounding effects of the digital divide across multiple areas of life.
- a) Explain how economic barriers to internet access can lead to reduced access to education.
- b) Discuss how reduced access to education can, in turn, worsen economic disparities in the next generation.
- c) Construct an argument for why the digital divide should be treated as a human rights issue rather than simply a technology access issue.
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Topic 7.4: Computing's Impact on Individuals and Society

1. Discuss the dual nature of computing's impact on individuals and society, evaluating both its benefits and its risks.
- a) Explain at least three positive impacts of computing on education, communication, and the economy, with specific examples from Pakistan.
- b) Explain at least three negative impacts of computing, including the digital divide, misinformation, and privacy concerns, with specific examples from Pakistan.

- c) Evaluate the claim that "technology itself is neither good nor bad — its impact depends on how it is designed, regulated, and used," using the example of social media's role in both the Peshawar attack awareness campaign and COVID-19 misinformation.
2. Construct a case study analysis of a real-world data privacy incident and its broader societal implications.
 - a) Summarize the NADRA data leakage incident as described in the text.
 - b) Analyze what this incident reveals about the risks of large-scale personal data collection.
 - c) Propose at least two policy or technical safeguards that could help prevent similar incidents in the future.
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Topic 7.5: Digital Citizenship and Ethical Considerations

1. Explain how critical thinking skills contribute to responsible information utilization. Provide examples of how these skills can be applied in real-life scenarios.
 - a) Define digital citizenship and responsible digital behavior.
 - b) Explain the ethical issues of plagiarism, copyright violation, and data bias, with an example of each.
 - c) Discuss how critical thinking helps a person avoid falling for misinformation or unethical data practices online.
 - d) Apply these principles to a real-life scenario: a student is asked to write a report using both online articles and an AI summarization tool. Explain how they should responsibly use these resources.
2. Discuss the importance of collaborative problem solving in a digital environment and provide examples of how human-machine collaboration can enhance this process.
 - a) Define human-machine collaboration and teamwork/knowledge sharing, with an example of each.
 - b) Explain the risks of over-relying on machine/AI output without applying human judgment.
 - c) Design a workflow for a group project that responsibly balances AI tool usage with human oversight and collaboration.
3. Design and justify a complete accessibility plan for a student-created website or digital report.
 - a) Explain the historical and ethical importance of accessible design, referencing how accessibility innovations often benefit a wider audience than originally intended.
 - b) Describe at least three specific accessibility techniques (e.g., ALT text, heading structure, color contrast, readable fonts) and how each should be implemented.
 - c) Explain how user feedback should be collected and used to continuously improve the accessibility of digital content.
 - d) Evaluate the statement: "Accessible design is not just for people with disabilities — it benefits everyone," using an example from outside computing (e.g., physical ramps) to support your argument.